

5 REASONS WHY YOU AGE

CELL DEPLETION This happens in important tissues, including the heart and the brain

UNWANTED CELLS Fat cells replace muscle, leading to diabetes and heart disease

CHROMOSOME MUTATION Telomeres, which cap chromosomes, decrease in length and cause decay

MITOCHONDRIA MUTATION Mitochondria, the tiny machines that power cells, can turn cancerous

JUNK BUILD-UP Waste hoards in cells and hardens arteries

10 DO'S AND DON'TS TO LIVE LONG

Don't oversleep: More than eight hours is not healthy

Be optimistic: It decreases the risk of early death by 50%

Lose weight: Obesity is linked to diabetes and heart disease

Get a pet: Pets reduce levels of the stress hormone, cortisol

Be rich: Higher incomes make it easy to engage in health-promoting behaviours

Quit smoking: It's been linked to most common causes of death in the elderly

Chill out: The anger-prone are six times more likely to have heart attacks by the age of 55

Pop antioxidants: Cinnamon to blueberries, they ward off cell damage and delay ageing

Marry well: Pick a spouse whose family shows longevity and it may help your kids too

Exercise a lot: Maintaining weight will give you energy and ward off stress

LONGEVITY AND MOOD

The Big Five personality traits that help people to live and thrive...

- Extroverted character
- Agreeable manner
- Conscientiousness
- Emotional stability
- Openness to experience



SKIN: Quit smoking. It causes premature wrinkles and ageing. Limit sun exposure, use sun-screen.



JOINTS: Try to strengthen those quadriceps to help prevent osteoarthritis, specially in the knees. To relieve pain and symp-

toms, apply heat to joints. There is a range of motion exercises that are good for the joints. But don't forget to warm up before beginning a work-out.

Sources: World Health Organisation, ICMR, JAMA, IOTF, International Osteoporosis Foundation, Indian Heart Journal, Harvard Medical JI.

lawyer with ITC, scoffs at the idea. But she may not have a choice. World demography has turned upside down and the super-old are thriving. "We are in the midst of a silent longevity revolution," says P.V. Ramamurti, the father of Indian gerontology. "Never before in human history has this kind of explosion happened." What's more, the oldest-old forms the fastest-growing segment among the seniors, adds demographer Ashish Bose, Director of the Society for Applied Research on Humanities, Delhi. The number of centenarians is growing significantly for the first time in history. "And this will redefine the concept of old age."

Statistics prove the point. More than half the world's oldest-old live in six countries:

China, US, India, Japan, Germany, and Russia. In 2000, 69 million people in the world formed the oldest-old. Among them, 7.8 million were 90-plus, and about 167,000 estimated to be 100 and beyond. According to the *UN Global Action on Ageing 2007*, they are growing at 4.4 per cent in India and China, compared to the world average of 2.6 per cent. In 1998, there were 1.3 million centenarians in the world, today there are 1.5 million in India alone, points out a HelpAge report. By 2050, the world's share of centenarians is projected to reach 3.3 million, led by China, US, Japan, and India.

Scientists are all fired up. And researchers—in stem cells, chromosomal telomeres, nanotechnology and more—are throwing up an alphabet soup of theories to explain longevity. Can organs be replaced the way we replace worn brakes on a car? Can genetic therapies rejuvenate the 'junk' we build up in our cells? Can microscopic nanobots be made to flow through the body, warning us of future health problems? Do stem cells hold the key? Dr Salim Yusuf, Director of Cardiology and Population Health Research Institute at McMaster University in Canada, compared the quest in a recent presentation at the All India Institute of Medical Sciences (AIIMS), Delhi, to the timeless questions raised by Paul Gauguin in his famous painting, *D'où Venons-Nous: "Where did we come from? What are we? Where are we going?"*. But beyond that philosophical big picture, some simple rules work. "How long you live depends partly on the genes you are born with and partly on lifestyle—how you live and eat, what types of stress and trauma you experience and what sort of a person you are," he asserts.

While the super-old are happy to offer homespun explanations for their longevity—"I never took a drink," "I never smoked," "I lived on oranges"—experts are trying to unravel the biological factors that allow some people to reach 100. Low fat and low protein diet also boost longevity, holds a

The number of the oldest-old in India is around

8m

They are growing at double the world average,

4.4%

every year